DES304: Emerging Technologies Stream
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Project partner: Empathic Computing Laboratory

Emerging Technologies & Speculative Design

Wero (The challenge)

How might we use emerging technologies and speculative design to reimagine how people in Aotearoa connect, collaborate, and create?

Introduction

This challenge invites students to critically speculate about emerging technologies and their possible socio-cultural implications, exploring how our ways of connecting, collaborating, and creating might be transformed in a plausible future Aotearoa.

As digital technologies become increasingly entangled with our everyday lives, and as AI continues to shape expectations about a new era of profound technological and social change, speculative design offers an approach for envisioning alternative futures to examine, question, and provoke discussion about how such change might affect human relationships and creative practices.

Working across one or more of three thematic lenses — *connect*, *collaborate*, and *create* — students will develop speculative artefacts, experiences, or systems that exist within a plausible future scenario in Aotearoa, or that critically reflect on what such a future might mean for people and communities.

Outcomes might include media artefacts from an imagined future Aotearoa (diegetic prototypes); installations, games, or interactive prototypes (including VR/AR, spatial computing, and wearable devices) that place an audience inside a speculative scenario; community decision-making tools; and reimagined public services or civic systems.

Keywords: Speculative Design, Emerging Technologies, Creativity, Collaboration, Social Innovation, Empathic Computing, Creative Computing, Digital Media, Augmented and Virtual Realities, AI, Game Design, Spatial Computing, Diegetic Prototyping, Provotypes.

What Will You Do?

Students will engage in a design-led inquiry that includes:

- **Background research**: Investigate global and local precedents in speculative design and emerging technology applications.
- **Identifying opportunity areas**: Define where design can open up critical questions or propose alternative possibilities: whether through speculative artefacts, experiences, devices, systems, or narrative scenarios.
- **Refining the scope**: Working within one or more of the thematic lenses — connect, collaborate, and create — narrow the challenge based on your insights (e.g., focus on a specific technology or a particular audience).
- **Methodology**: Adopt and justify an appropriate design methodology to investigate, ideate, prototype, and test your intervention.
- **Prototyping and communication**
    - At the end of S1, come up with 3 concept solutions.

- During S2, refine the best solution and develop it. Develop a concept that communicates value clearly and meaningfully through highly curated design communication.

- **Defining impact (semester 2)**: Articulate how your intervention provokes reflection, invites debate, or opens up alternative ways of thinking about future socio-technical shifts in Aotearoa.

Methodological Scope

Students are encouraged to select approaches that resonate with their values, experiences, and sense of place. You should define a methodological framework to guide your inquiry, while experimenting with a range of methods drawn from within or across different methodologies. You may adapt or combine established methodologies — for example, by drawing on methods typically associated with one framework while working within another, or by integrating conceptual aspects of different approaches.

Adopt a methodology that enables you to:

- Frame and reframe the design challenge
- Explore new perspectives
- Develop and prototype a concept solution
- Plan for real impact

Within the broader field of speculative design, there are multiple established frameworks that students might draw on, including critical design, design fiction, more-than-human centred design, and indigenous futurisms.

Justify your approach with reference to relevant literature, precedents, and other supporting evidence and insights.

Focus Areas, Opportunities, and Resources

Students may choose to engage with one or more of the following thematic lenses:

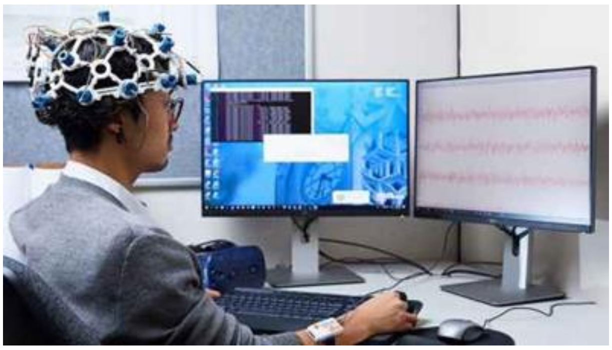
- **Connect:** How might emerging technologies reshape the ways people in Aotearoa relate to one another, to place, or to shared histories and identities?

- **Collaborate**: How might emerging technologies enable new ways of working together, or new forms of shared decision-making or co-creation, in Aotearoa?

- **Create:** How might emerging technologies transform making practices, or understandings of creativity, for practitioners and creative communities in Aotearoa?

Students in this stream will develop relationships with relevant experts from across the University, and will have access to equipment and facilities from the Design Lab, Engineering's Smart Digital Lab, the Digital Research Hub, and the Empathic Computing Laboratory. Available equipment includes high-performance computers with GPUs suitable for AI applications, projectors, RGBD sensors, 360° video cameras, Oculus Quest VR headsets, and Hololens 2 AR headsets.

About Our Project Partner: The Empathic Computing Laboratory



The Empathic Computing Laboratory is a research group at the Auckland Bioengineering Institute with a focus on using emerging technologies to create shared understanding between people. It is led by world-renowned augmented reality expert Professor Mark Billinghurst.

An example of technology with an empathic element is a headset that has the capacity to relay emotional and physiological information about the wearer, such as their facial expression and heart-rate. Facial expression technology currently exists in AffectiveWear, glass with small photo-sensors which measure the distance from the glass frame to the skin. These measurements change when we smile, frown or gasp.

“We wanted to relate the cues you usually have in face-to-face conversation, as these show understanding. Using our technology, it all becomes a much more immersive experience. You feel as though you’re standing inside the body of the person wearing the headset… In trials of sharing eye gaze, we’ve found people experience a stronger sense of collaboration and communication,” says Professor Mark Billinghurst.

The work conducted in the Empathic Computing Lab is at the junction of three computer interface trends. These trends are:

- The way we capture content, which has advanced from still photography in the 1850s to today’s streaming 360-degree video on a portable machine.
- Increasing bandwidth, which allows you to download a movie in seconds and do higher-quality video conferencing.
- ‘Implicit understanding’ where computers are able to watch and listen to us in order to understand what we are doing e.g. being able to play games on a console by moving your body around. Professor Mark Billinghurst explains: “So, to some extent, computers have become more human-like. They can recognise our behaviour”.

Useful Resources

The Empathic Computing Lab (UoA): https://empathiccomputing.org/
The Smart Lab (UoA): https://www.sdl.auckland.ac.nz/
Design Lab (UoA): https://designtech.blogs.auckland.ac.nz/
The Digital Research Hub (UoA): https://www.drh.nz/